

- 4 Fig. 4.1 is a diagram showing the transcription of part of the *COL1A2* gene that codes for a collagen polypeptide. The part of the *COL1A2* gene shown is a section of exon. Structure **A** represents an enzyme involved in transcription.

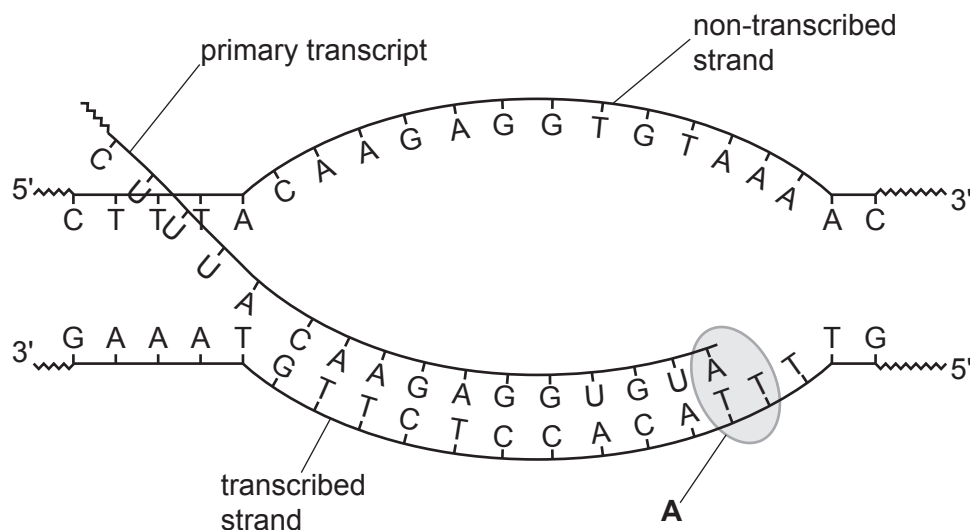


Fig. 4.1

- (a) (i) Name the enzyme labelled **A** in Fig. 4.1.
 [1]
- (ii) Name the bond that forms between the nucleotides in the primary transcript.
 [1]
- (b) (i) State the number of amino acids that are coded for by the sequence of nucleotides on the primary transcript shown in Fig. 4.1.
 [1]
- (ii) Use the information in Fig. 4.1 to explain why one of the strands of DNA is **not** transcribed.

 [2]

- (c) New mitochondria are formed when a mitochondrion divides into two.

Fig. 4.2 is a transmission electron micrograph of a mitochondrion that is dividing.

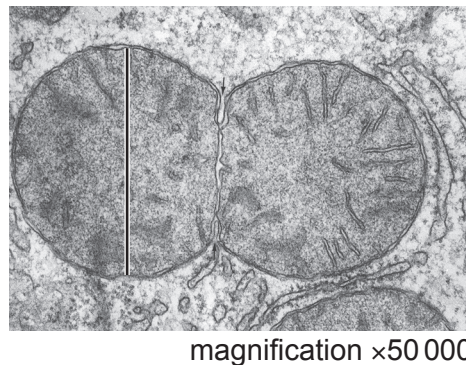


Fig. 4.2

- (i) State **two** features of mitochondria that are **visible** in Fig. 4.2.

1

2 [2]

- (ii) The line on Fig. 4.2 shows the diameter of one of the mitochondria.

Calculate the actual diameter of the mitochondrion.

Give your answer to one significant figure.

answer = μm [1]

- (d) Mitochondrial DNA codes for some polypeptides of proteins used within the mitochondrion. Some of the proteins allow movement of ions into and out of the mitochondria.

Outline the ways in which ions can move into mitochondria.

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..... [3]

[Total: 11]